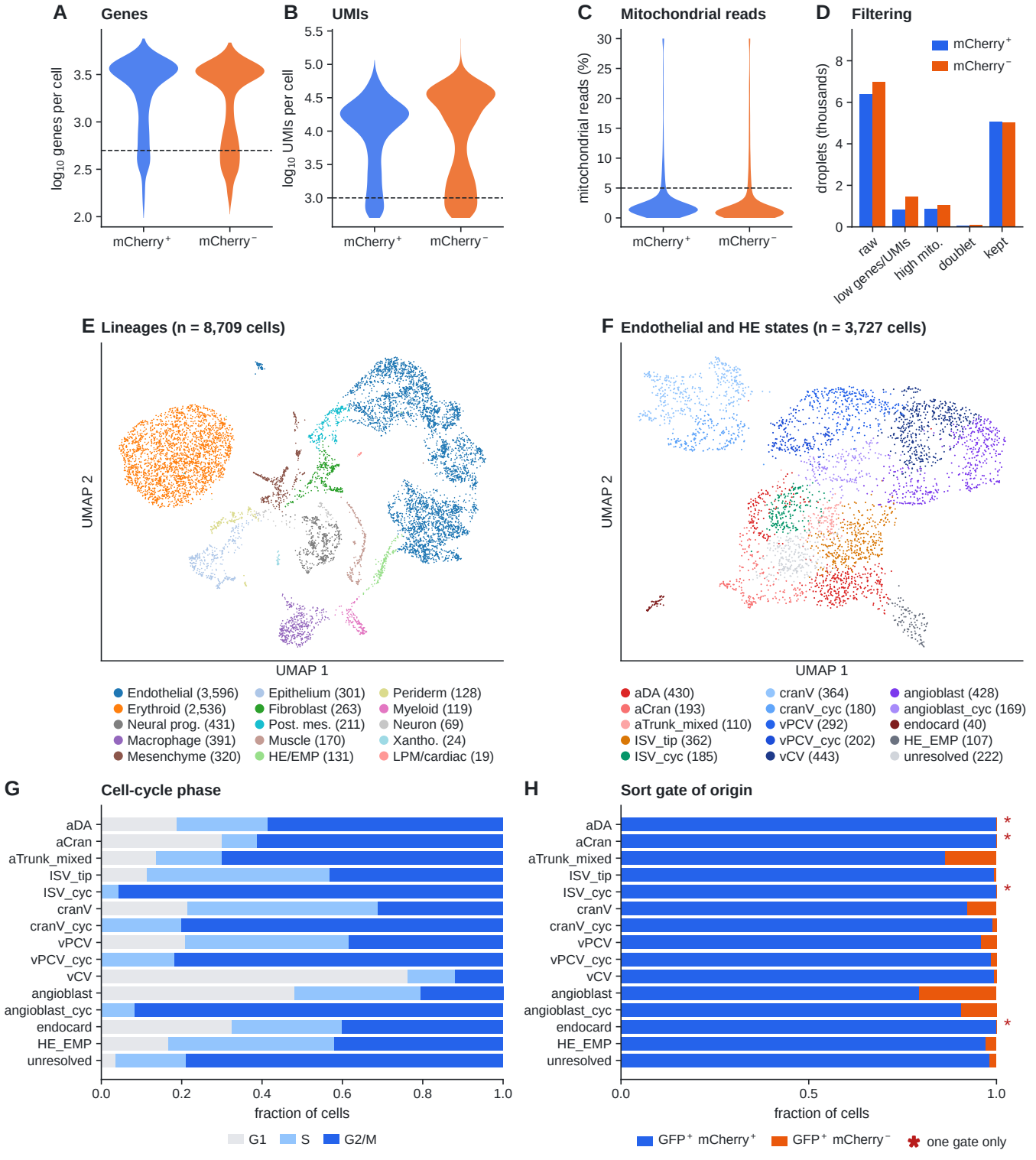

A
Genes
log10 genes per cell
3.5
3.0
2.5
2.0
mCherry+
mCherry−
B
UMIs
log10 UMIs per cell
5.5
5.0
4.5
4.0
3.5
3.0
mCherry+
mCherry−
C
Mitochondrial reads
mitochondrial reads (%)
30
25
20
15
10
5
0
mCherry+
mCherry−
D
Filtering
droplets (thousands)
8
6
4
2
0
mCherry+
mCherry−
raw
low genes/UMIs
high mito.
doublet
kept
E
Lineages (n = 8,709 cells)
UMAP 2
UMAP 1
Endothelial (3,596)
Erythroid (2,536)
Neural prog. (431)
Macrophage (391)
Mesenchyme (320)
Epithelium (301)
Fibroblast (263)
Post. mes. (211)
Muscle (170)
HE/EMP (131)
Periderm (128)
Myeloid (119)
Neuron (69)
Xantho. (24)
LPM/cardiac (19)
F
Endothelial and HE states (n = 3,727 cells)
UMAP 2
UMAP 1
aDA (430)
aCran (193)
aTrunk_mixed (110)
ISV_tip (362)
ISV_cyc (185)
cranV (364)
cranV_cyc (180)
vPCV (292)
vPCV_cyc (202)
vCV (443)
angioblast (428)
angioblast_cyc (169)
endocard (40)
HE_EMP (107)
unresolved (222)
G
Cell-cycle phase
aDA
aCran
aTrunk_mixed
ISV_tip
ISV_cyc
cranV
cranV_cyc
vPCV
vPCV_cyc
vCV
angioblast
angioblast_cyc
endocard
HE_EMP
unresolved
0.0
0.2
0.4
0.6
0.8
1.0
fraction of cells
G1
S
G2/M
H
Sort gate of origin
aDA
aCran
aTrunk_mixed
ISV_tip
ISV_cyc
cranV
cranV_cyc
vPCV
vPCV_cyc
vCV
angioblast
angioblast_cyc
endocard
HE_EMP
unresolved
*
0.0
0.5
1.0
fraction of cells
GFP+ mCherry+
GFP+ mCherry−
one gate only